

measurements — in this case, the values in the arrays x and y . The function returns, in addition to other data, the slope and intercept of the regression line that is stored in the variables m and c , respectively. Line 10 uses the values of slope and intercept to generate the regression line. Line 11 plots the regression line in green. Finally, Line 12 displays all the plots. Figure 9 shows the image generated on execution of the program. The data points are displayed in yellow and the regression line is shown in green.

In the last three articles in this series, we have tried to learn more and more concepts from probability and statistics. Though the treatment of the subject is in no way comprehensive, I think we have arrived at a good starting point and it is time to shift our focus to other matters of equal importance.

Introduction to Keras

Now, let us learn to use Keras. Keras is used along with TensorFlow. Hence, let us begin by exploring TensorFlow a bit more. Consider the program shown in Figure 10. Though this program contains just four lines of code, this is our first encounter with neural networks in this series. Now, let us try to understand the working of the program. Line 1 imports the library TensorFlow. Line 2 creates a tensor called x . Lines 3 and 4 apply two activation functions called ReLU (Rectified Linear Unit) and Leaky ReLU, respectively, on the tensor x . In neural networks, the activation function of a node defines the output of that node given an input or set of inputs. ReLU activation function is an activation function defined as the positive part of its argument. The output of the code in Line 3 is shown in Figure 10. It can be observed that the negative values in the tensor x are replaced with zeros after applying the ReLU activation function. Leaky ReLU activation function is a modified

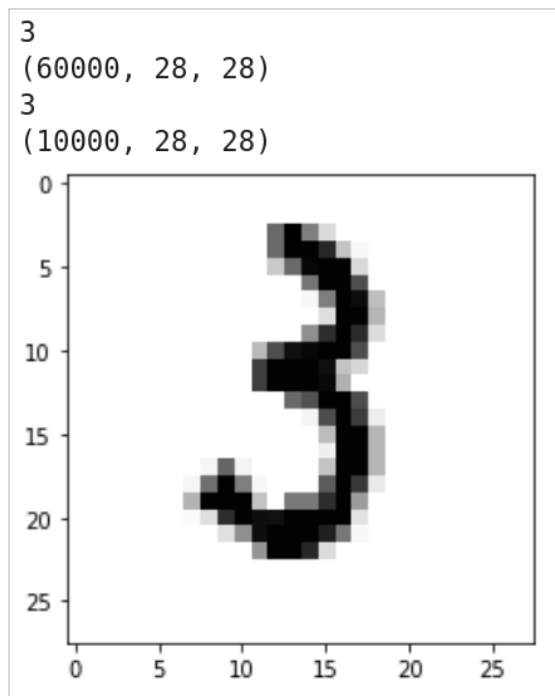


Figure 12: A sample from the data set

version of the ReLU activation function. From the output of the code in Line 4 shown in Figure 10, it can be verified that positive values are retained as such, and 20 per cent of the negative values are retained on applying the Leaky ReLU activation function. Though we are moving on to discuss Keras, we will revisit this topic in the future to learn more about neural networks and activation functions.

Now, let us start using Keras. The installation of Keras can also be done very easily with Anaconda Navigator. Consider the program shown in Figure 11. We will use our first data set in this program. For the time being, we will only import and display sample data from the data set. But in the next article, we will use the same data set for training and testing our model, thus beginning

the next phase in our journey to develop AI and machine learning based applications.

Now, let us try to understand the working of the program. Line 1 imports Keras. Line 2 imports the MNIST database of handwritten digits. It has a training data set of 60,000 samples and a test data set of 10,000 samples. Line 3 imports Matplotlib. Line 5 loads the data set. Lines 6 to 9 print the dimension and shape of the training data and the test data. Figure 12 shows the output of these lines

of code. It can be seen that both the training data and the test data are three-dimensional. It can also be verified that there are 60,000 training images of 28 x 28 pixel quality and 10,000 test images of 28 x 28 pixel quality. Line 10 loads the 1234th training image. Lines 11 and 12 display this image. Figure 12 shows this handwritten image of the digit 3.

Now, it is time to wind-up our discussion. We are at the halfway mark of our journey through AI and machine learning. We have started discussing neural networks, a discussion we will continue in the next article. We have also come across our first data set with Keras. In the next article, we will continue discussing the use of Keras by training with this data set. We will also have our first interaction with scikit-learn, another powerful Python library for machine learning. **END** 🐧

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